BeepBoop at the TechHub: A Case Study of a Low-Cost LLM-Driven Social Robot for University Helpdesk Support

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Abstract. Frontline university helpdesks often face high volumes of routine enquiries. Deploying social robots in this context allows university staff to focus on more complex requests, while the robots can provide an on-demand source of assistance with social interaction capabilities. This paper introduces a case study of a low-cost LLM-driven social robot co-designed with the university iSolutions team to support helpdesk service operations. The proposed conversational framework integrates retrieval-augmented generation to ensure domain relevance and incorporates information from university resources. The system also maintains short-term dialogue memory to support short interaction windows and multi-turn exchanges. Evaluations were conducted to assess the robot's question-answering performance. It was followed by an experiment at the Tech Hub involving 40 participants to examine users' perceptions of the robot's usability and perceived usefulness in a university helpdesk setting.

Keywords: human-robot interaction · social robot · large language models · helpdesk support

1 Introduction

University helpdesks frequently receive high volumes of enquiries, especially at the beginning of the semester or during examination periods [36]. To maintain accessible and supportive services for students during these peak times, it is beneficial to provide first-line assistance capable of handling large numbers of repetitive queries without diverting staff from more complex cases. Traditional search engines are not well designed to effectively filter user requests in such contexts, meaning that students often spend considerable time and effort searching for the necessary information. Furthermore, information related to admissions, modules, the student hub, and IT services is often distributed across multiple

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websites and portals, making it difficult for students to synthesise relevant and updated information. Recent advances in Large Language Models (LLMs) offer a promising opportunity to address this challenge by enabling the extraction of structured knowledge from unstructured text and supporting domain-specific reasoning based on university data resources [23,19].

However, deploying LLM-based chatbot systems in public-facing service roles introduces interaction constraints due to the limited social engagement typically present in screen-based conversations, compared to the way students interact with a traditional helpdesk staff. To address this limitation, the use of social robots in service environments has been widely investigated, ranging from robots for Nescafe machines to service robots deployed in chocolate stores, shopping malls, and bakery shops [24,31,35,18]. Compared to a purely text-based chatbot, a robot's physical embodiment, together with its verbal and non-verbal cues, plays an important role in attracting users and enhancing both the frequency and duration of interactions [14,9].

Motivated by this context, this research, conducted in collaboration with the iSolutions team at the University of Southampton, aims to explore a low-cost and scalable social robot that can be deployed across Tech hubs within the university as a first-line assistance system. The robot could be designed to handle repetitive queries while providing socially interactive support for students.

While LLM-based social robots have been widely studied in different personal use and well-being settings, the use of LLM-based robots in university environments remains an emerging research topic, particularly studies on low-cost and scalable social robots designed for helpdesk services. The primary contributions of this study are as follows:

- We introduce a low-cost social robot platform co-designed with our stakeholders to support deployment in university service environments.
- We propose a helpdesk conversational framework that grounds answers in university documentation, maintains short-term context, and pairs responses with robot interaction cues.
- We present a case study of validating the robot for helpdesk support through a field experiment at a UK university Tech hub. Quantitative and qualitative evaluations were conducted to assess the proposed approach. The robot hardware design, conversational framework, and dataset tailored to the university service setting will be made publicly available to the HRI community, enabling further studies on LLM-based social robots in this domain.

2 Related Work

2.1 LLM-based Chatbots for University Q&A

University question answering (Q&A) services have increasingly shifted from rule-based chatbots toward LLM-driven virtual assistants, which can interpret open-ended queries and generate fluent responses. In the university context, Neupane et al. [23] present BarkPlug v.2, an LLM-based chatbot for campus

resources that illustrates both the practical value of conversational access to institutional information and the importance of maintaining usability in realistic deployments. Similarly, in [1], the authors introduce an LLM-based approach for academic advising in higher education, in which the model is fine-tuned using information collected from university resources. Rather than relying on fine-tuning, a common strategy for improving reliability and adaptability in such systems is retrieval-augmented generation (RAG), where responses are conditioned on retrieved documentation that can dynamically update the system's knowledge, rather than relying solely on the model's parametric knowledge [19].

Despite their potential, LLM-based chatbots are typically deployed as web interfaces or tablet kiosks, which do not address the need for social engagement at helpdesk points where services are traditionally delivered face-to-face by staff. In such settings, users expect clear turn-taking and backchannel feedback indicating that the system is processing their request. When these social cues are absent, delays in system responses can be misinterpreted as breakdowns relative to human conversational timing norms [27,30]. Moreover, compared to virtual AI agents or tablet kiosks, socially embodied agents have been shown to be more effective in attracting users and increasing both the frequency and duration of interactions [14,9]. This motivates complementing university Q&A chatbots with interactive agents capable of providing social and multimodal interaction mechanisms that more closely resemble a traditional helpdesk experience.

2.2 Social Robots in Public Environments

Substantial HRI literature demonstrates that robot embodiment can attract attention, structure interaction, and sustain engagement in ways that screen-based systems cannot. In retail and food service settings, Niemelä et al. [24] identify both the perceived value of an embodied presence (e.g., welcoming, guidance, and atmosphere) and its practical constraints (e.g., reliability, noise, and role clarity). Similarly, Song et al. [31] and Nguyen et al. [35] show that interaction style and coordinated non-verbal behaviours shape perceived helpfulness and empathy in service contexts. A field study conducted in café settings further confirms that user acceptance of robots is tightly coupled with role clarity and the legibility of robot behaviour [18].

The rapid advancement of LLMs in recent years has created new opportunities to integrate them into robotic dialogue systems, offering a promising direction for improving the fluency, adaptability, and contextual awareness of robot speech [37]. In [32], the authors introduce an LLM-based Human–Robot Interaction (HRI) system called VITA, which enables a social robot to provide mental well-being support to users through natural conversational interaction. Similarly, Cao et al. [7] demonstrate that incorporating Retrieval-Augmented Generation (RAG) can enhance an LLM's ability to retrieve relevant user personality traits, thereby producing more contextually appropriate responses and improving the engagement of HRI interactions. Similarly, in [34], the authors propose SYNERGY, an LLM-based framework integrated with RAG to support daily social communication and assistance for users in smart-home environments.

While recent studies have explored LLM-based social robots in personal and well-being contexts [32,7,34], the deployment of such systems in university service environments remains an emerging research topic. University helpdesk scenarios introduce distinct interaction and operation challenges, including high user volume, short interaction windows, and the need to provide accurate and updated university-related information while maintaining social engagement comparable to interactions with a staff member. This paper presents a case study on an LLM-based social robot designed for university helpdesk support. The study explores how such a system can operate in a high-traffic service environment and investigates users' perceptions of the robot's usability and perceived usefulness.

3 Robot Design

The robot, named BeepBoop, was co-designed with the iSolutions team at our university with the goal of developing a low-cost social robot platform that can be easily maintained by university staff, including those with limited robotics experience. The system is intended to support effective interactions within short time windows in high-traffic service environments. To address these requirements, the robot design was inspired by five principles commonly emphasised in HRI and service robotics literature. The following sections discuss these principles in more detail.

3.1 Appearance and Approachability

The robot's physical form was designed to encourage user engagement. Standing approximately 60 centimetres in height, the table-mounted configuration reduces perceived physical dominance and lowers barriers to close-range interaction compared to full-scale humanoid platforms. This compact design also enables flexible deployment across different tech hubs within the university.

To avoid the uncanny valley effect [22], the robot's appearance is intentionally stylised rather than realistic. The design incorporates minimal anthropomorphic cues, consisting of a screen-based face and simple arms. This helps prevent users from overestimating the robot's intelligence and supports appropriate expectation calibration [10].

Additionally, the robot adopts a smooth, rounded profile (Fig. 1). By eliminating sharp edges, the design leverages the human preference for curved, non-threatening visual forms [2], reinforcing a safe and approachable presence for interaction. This design approach also aligns with findings that university students tend to prefer non-threatening, machine-like educational robots [26].

3.2 Safety and Robustness

Designed for deployment in high-traffic university helpdesk environments, the robot's mechanical architecture prioritises both physical safety and operational reliability. The design of a wide footprint and low centre of gravity aims to reduce

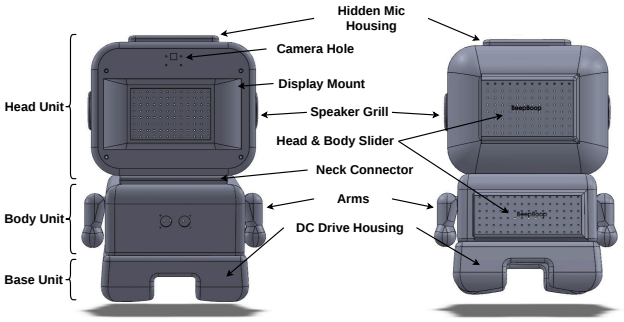


Fig. 1: Annotated CAD model of the robot, illustrating the modular structural partitioning.

the risk of toppling during arm movements or incidental contact. The robot's smooth and rounded contours further minimise potential pinch hazards while reinforcing perceptual safety cues associated with non-threatening shapes [2].

To ensure sustained reliability during operation, the platform employs a dual-rail power architecture [11]. By isolating high-current servo loads from the robot's main board power rail, the design reduces electrical noise, prevents audio interference, and avoids transient compute resets during operation.

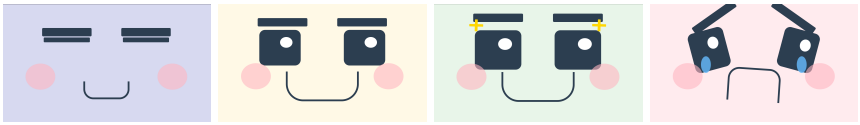
3.3 Affordability and Manufacturability

To support low-cost manufacturability and rapid iterative prototyping, the robot uses a screen-rendered face to convey facial expressions. This approach avoids the mechanical complexity and potential failure points associated with actuated expressive faces used in other social robots such as Kismet or Kaspar [17].

The robot employs a modular, 3D-printed chassis powered by a Raspberry Pi for the on-device user interface, data processing, and networking without the cost of integrating a PC motherboard. Following the hardware architecture of the JARI social robot [21], the robot main board manages high-level interaction processes while delegating low-level motion control to dedicated microcontrollers. This decoupled computing architecture improves system stability and responsiveness during long-term operation.

3.4 Operational Autonomy and Maintainability

To support long-term deployment at the Tech Hub, the robot was designed with a focus on operational endurance and ease of maintenance. As presented in Fig. 1, the robot adopts a modular approach that allows individual components to be easily accessed and replaced. This design enables university staff without a robotics background to perform basic maintenance and repairs on the platform. This approach aligns with open-source robotics design principles, where



(a) Sleeping (b) Idle (c) Listening (d) Error

Fig. 2: Examples of the robot facial expressions.

modularity and rapid repair are essential for sustaining operation in real-world settings [33].

3.5 Interactivity

Interactivity is a key feature of the BeepBoop robot, enabling it to communicate with users in a neutral and engaging manner. Prior research has shown that non-verbal social cues play an important role in human–robot interaction [5,35]. In particular, gestures can serve as a secondary signalling layer that makes internal state changes legible even when the robot's face is not directly visible or when audio is masked by ambient noise. Accordingly, the robot's arms provide motion cues that can be synchronised with the robot's conversational state.

The robot's facial display is synchronised with its speech to support multimodal communication. This capability is particularly important in noisy and crowded service environments, where users may rely on multiple communication channels to interpret the robot's messages [35]. The robot's internal states (e.g., idle, listening, error) are also displayed on the screen face, as shown in Fig. 2. To ensure that expressions remain easily interpretable while avoiding human-like facial realism, the design adopts highly rated robot facial attributes identified in [17].

User perception of service robots depends not only on task completion but also on the robot's ability to provide a warm and engaging interaction [24]. To support this, the BeepBoop robot incorporates active speaker tracking, enabling it to orient toward the current speaking user and thereby strengthen perceived interaction naturalness [29]. In addition, several vision-based interactive mini-games were included to engage passers-by who approach out of curiosity or while waiting in service queues to speak with a member of staff.

4 LLM-driven Conversational Framework

Figure 3 illustrates the proposed conversational framework. The framework is designed to address the constraints of a public helpdesk environment, where interactions are typically brief, follow-up questions are common, and system processing latency must be presented to users in a transparent way.

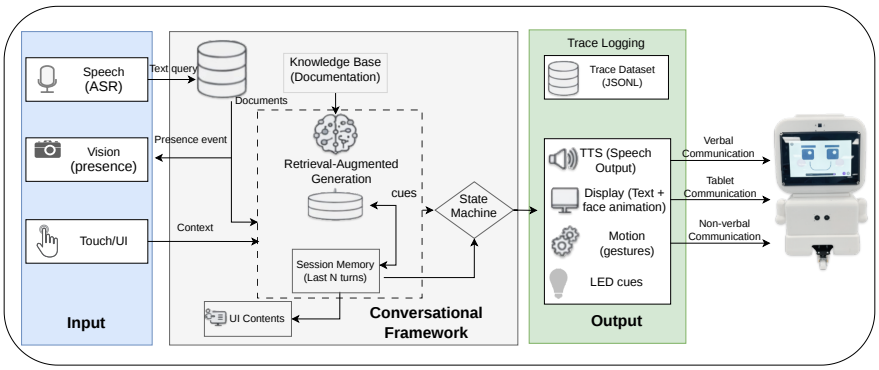


Fig. 3: Conversational framework architecture. Multimodal inputs are routed through interaction-state control and retrieval-augmented generation with session memory, producing multimodal outputs displayed on a social robot.

4.1 Multimodal Inputs

As presented in Fig. 3, user speech is first transcribed into text, forming the primary query for the system. Vision-based presence detection provides lightweight engagement cues, such as user tracking, to support interaction initiation and state transitions. The robot's tablet interface offers an alternative input channel for receiving textual inputs or touching events, improving the system robustness in public interaction environments.

4.2 Knowledge Base

The knowledge base aggregates information related to university services, allowing the system to access dynamically updated content. In our setting, the sources include the iSolutions online knowledge base, selected ECS faculty intranet pages, and the University of Southampton Open Data Service. The documents are normalised to remove formatting artefacts and segmented into semantically coherent chunks suitable for embedding-based retrieval. This representation enables efficient similarity search while preserving procedural coherence for multi-step support instructions.

4.3 Retrieval-Augmented Generation (RAG)

At query time, the system performs similarity-based retrieval over embedded documentation chunks and inserts the retrieved context into a structured prompt template, as presented by the RAG module in Fig. 3. This grounding step helps reduce unsupported responses by conditioning generation on verified university sources [19]. When the retrieved context is insufficient or poorly matched to the query, our prompting design encourages conservative behaviour, such as asking

Algorithm 1 Session transcript (UI) and lightweight memory for prompt context

Require: Query q_t; UI transcript $U_t = \langle (s_i, m_i) \rangle$; assistant-memory $H_t = \langle a_i \rangle$; max turns N; prompt template $\mathcal{P}$; system instruction S; retrieved context c_t
Ensure: Answer a_t; updated U_{t+1} and H_{t+1}

if ClearHistory invoked **or** session restarts **then**
  $U_t \leftarrow \emptyset$; $H_t \leftarrow \emptyset$
end if
$U_t \leftarrow U_t \,\|\, \langle (\texttt{user}, q_t) \rangle$ ▷ UI stores user turn for display
$h_t \leftarrow \textsc{Format}(H_t)$ ▷ lightweight memory used as history field
$X_t \leftarrow \{\texttt{system} = S,\ \texttt{question} = q_t,\ \texttt{context} = c_t,\ \texttt{history} = h_t\}$
$a_t \leftarrow \mathcal{P}(X_t)$
$U_{t+1} \leftarrow U_t \,\|\, \langle (\texttt{bot}, a_t) \rangle$ ▷ UI stores bot reply
$H'_t \leftarrow H_t \,\|\, \langle a_t \rangle$ ▷ MemoryHandler stores bot reply
if $|H'_t| > N$ **then**
  $H_{t+1} \leftarrow \textsc{Tail}(H'_t, N)$
else
  $H_{t+1} \leftarrow H'_t$
end if
return $a_t,\ U_{t+1},\ H_{t+1}$

clarifying questions or signalling uncertainty, rather than producing speculative instructions.

4.4 Session Memory and UI Contents

To support natural multi-turn interaction, the framework maintains a bounded session memory as presented by the Session Memory module in Fig. 3. Recent dialogue turns are retained within a fixed window to provide context for underspecified follow-ups (e.g., pronouns or elliptical references), without enabling long-term personalisation. The session context is also presented to users through the User Interface (UI) and can be explicitly cleared to ensure a clean boundary between users in a public setting. Algorithm 1 specifies the session-bounded memory mechanism used to construct prompt context.

4.5 Interaction-State Control and Multimodal Outputs

Interaction is coordinated by a deterministic finite state machine (FSM) with seven states: *Sleep*, *Idle*, *Listening*, *Thinking*, *Speaking*, *Error*, and *Recover*, as shown in Fig. 3. This design is motivated by evidence that the time required to generate LLM responses often exceeds typical human turn-taking norms [27,30], making interaction-state legibility critical in public-facing settings.

In our designed conversational framework, a deterministic FSM was chosen over asynchronous event-driven control to reduce behavioural jitter in high-traffic

environments, where ambient noise and incidental movement could otherwise trigger unintended state transitions. The FSM also generates multimodal outputs displayed on the robot, including both verbal and non-verbal channels. This coupling between interaction states and output channels ensures users receive immediate feedback during listening and processing, helping maintain turn-taking continuity despite system latency.

5 Question Answering Evaluation

To evaluate the effectiveness of the proposed Q&A approach, the following section describes the data preprocessing steps used to construct the robot's knowledge base and the evaluation methodology used to assess the performance of the proposed LLM-based system across different metrics.

5.1 Knowledge Base Construction

Data collection and cleaning began by compiling a list of official URLs covering common helpdesk topics, supplemented with "frequently asked questions" used by university helpdesk staff. Each page was retrieved and converted to plain text by removing navigation elements and other non-informative content. The resulting text was then normalised and cleaned to eliminate redundant sections.

During data segmentation and metadata enrichment, documents were divided into semantically coherent chunks to support passage-level retrieval. Chunk boundaries were aligned with headings and paragraph structures where possible, with a small overlap between neighbouring chunks to reduce boundary effects for multi-step instructions. Each chunk was stored with metadata such as the source URL, page title, section heading path, and document identifier, enabling traceability in the UI and facilitating offline inspection of retrieved evidence.

Finally, the chunks were embedded and indexed in a vector store for similarity-based retrieval at inference time. This representation allows the robot to retrieve a compact set of relevant passages, which are then provided to the LLM during answer generation. The same preprocessing and indexing pipeline can be re-run to refresh the knowledge base when source pages change, while maintaining a consistent retrieval interface for the LLM across experiments.

5.2 Evaluation Metrics

The following metrics are used to validate the generated answer and the retrieved context across different metrics commonly used in this domain [12,23]. Here, (↑) indicates that higher values correspond to better performance.

- **BLEU(↑)**: n-gram overlap between the generated and the reference answer.
- **chrF(↑)**: n-gram F-score between the generated and the reference answer.
- **ROUGE(↑)** : how much reference content is covered by the generated answer.

- **Semantic Similarity (Semantic Sim)(↑)**: similarity between the generated and reference answers on the text embedding space.
- **Answer Relevancy (Answer Relev)(↑)**: how relevant the generated answer addresses the user's question.
- **Answer Accuracy (Answer Acc)(↑)**: correctness of the generated answer with respect to the reference answer.
- **Faithfulness(↑)**: how factually consistent the response is with the retrieved context.
- **Context precision(↑)**: the proportion of the retrieved context that is relevant to answering the question.
- **Context recall(↑)**: how effectively a model retrieves all relevant pieces of information.
- **Context F1(↑)**: mean of context precision and context recall.

5.3 Results and Discussion

The experiment was conducted using 135 independent prompts, consisting of 5 context-retrieval trials for 27 manually-written pairs of questions and reference answers to simulate queries that may be commonly handled by the university helpdesk service team. The Gemini-3-Flash model was used as the backbone of the conversational framework. It was chosen due to its reliable uptime, low API cost and low response latency, essential for sustaining user immersion and reducing usage friction. A local LLM was briefly considered, but deemed implausible with Pi's low hardware specifications. To evaluate the efficiency of the RAG module, the system was tested under two configurations: the conversational framework with RAG (*w-RAG*) and without RAG (*w/o-RAG*). For each query, we recorded the model's generated response and the corresponding answer. For the *w-RAG* configuration, the retrieved context passages were also logged. Table 1 summarises the evaluation results for both *w-RAG* and *w/o-RAG* models using metrics that compare generated responses with reference answers. Table 2 presents evaluation results that assess the *w-RAG* model's responses with respect to the retrieved context.

Model	**NLP Comparison**				**LLM-as-a-judge**	
	BLEU	**chrF**	**ROUGE**	**Semantic Sim.**	**Answer Relev.**	**Answer Acc.**
w/o-RAG	0.0663	0.2976	0.1926	0.8279	0.7594	0.1148
w-RAG	**0.1562**	**0.4064**	**0.3150**	**0.9131**	**0.7636**	**0.4389**

Table 1: Comparison of *w/o-RAG* and the *w-RAG* on response-based metrics.

The experimental results in Table 1 show that the *w-RAG* outperforms *w/o-RAG* across all response-based evaluation metrics. For lexical overlap, the w-RAG configuration significantly improves BLEU (0.1562 vs. 0.0663), chrF

(0.4064 vs. 0.2976), and ROUGE (0.3150 vs. 0.1926), indicating closer alignment between the generated answers and the reference responses in the textual space. The improvement is even more evident in semantic-oriented metrics. The answer accuracy increases from 0.1148 to 0.4389. Although this score may appear low, achieving a high metric is unlikely due to metric strictness and limitations of the evaluator LLM, so emphasis is placed on the $\approx$ 4x increase. The semantic similarity rises from 0.8279 to 0.9131. These results suggest that retrieved context helps improve both factual correctness and semantic alignment with the ground-truth answers. The answer relevance remains high in both model configurations (0.7594 w/o-RAG vs. 0.7636 w-RAG), indicating that the base LLM can generally produce on-topic responses. On the other hand, RAG plays a crucial role in improving factual correctness, thereby better supporting the Q&A requirements of the helpdesk service environment.

	Faithfulness	**Context Precision**	**Context Recall**	**Context F1**
w-RAG	0.8691	0.8444	0.7572	0.7481

Table 2: Performance of *w-RAG* model on context-based metrics

Table 2 reports context-based metrics that evaluate the generated responses with respect to the retrieved context. The faithfulness score of 0.8691 indicates that most generated answers are well supported by the retrieved evidence rather than containing unsupported claims. Retrieval precision is also high (0.8444), suggesting that most retrieved passages are relevant to the query. However, the context recall is slightly lower (0.7572), indicating that some queries may require additional or more targeted evidence than what is consistently retrieved. The resulting context F1 score of 0.7481 suggests that although the pipeline generally retrieves relevant supporting context, there is still room to improve coverage for more complex or multi-step helpdesk queries.

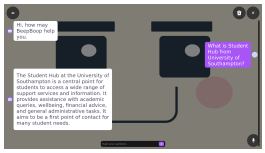
6 Operation at the Techhub

6.1 Interaction Scenario

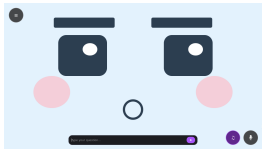
A field experiment was conducted at the iSolutions Tech Hub in the University of Southampton, one of the campus's busiest helpdesk points. The study was carried out over two weeks with a total of 40 participants ($n = 40$ users), including 26 students, 12 staff members, and 2 visitors, recruited under university ethics approval. Prior to the experiment, participants were briefed on the study's purpose and encouraged to explore the robot's functionalities. During the experiment, each participant was asked to explore all the robot's functions while an assistant remained nearby to provide help if needed, without interfering in the interaction. Over the two-week deployment at the Tech Hub, a total of 169 distinct query–response exchanges were collected from the 40 participants.



(a) Human-Robot Interaction



(b) Tablet Comm.



(c) Verbal Comm.



(d) Facial and Gesture Comm.

Fig. 4: (a) An example interaction scenario at iSolutions Tech Hub comprising of multimodal communication; (b) Tablet communication; (c) Verbal communication; (d) Facial and gesture communication.

6.2 Subjective Evaluation

After completing the HRI experiment, a subjective evaluation was conducted to understand users' interaction experiences. Participants completed a questionnaire using a 5-point Likert scale. First, they filled in the System Usability Scale (SUS) [6], a widely used instrument for quantifying perceived system usability, consisting of 10 items [28]. Drawing on validated HRI measurement tools such as the Godspeed Questionnaire [3] and the Robotic Social Attributes Scale (RoSAS) [8], we also evaluated the robot's perceived persona, including *Friendliness, Cuteness, Lifelikeness*, and *Intelligence*. In addition, participants assessed their experience with key functions of the robot, including *Speech Understanding, Response Speed, Voice Audibility, Text Readability, Motion Fluidity*, and *Camera Detection*. This evaluation approach allows us to systematically identify potential functional bottlenecks in the robot system, a methodology that has proven effective in recent evaluations of embodied LLM agents [37]. Finally, participants were asked to rate their overall satisfaction with the robot's Q&A capability and to rank their favourite aspects when interacting with the robot.

6.3 Latency in the Robot Q&A

Using the system logs that collected during the experiment at the Tech Hub, we first measured response latency, defined as the time from the cessation of a user's spoken query to the initiation of the robot's Text-to-Speech (TTS) response. Analysing the $n = 169$ logged query-response exchanges reveals a right-skewed latency distribution as presented in Fig. 5. Across all interaction sessions, the system exhibited a mean latency of 5.45 s and a median of 3.31 s.

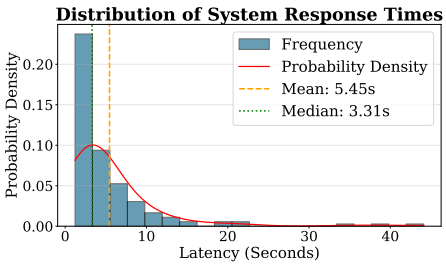


Fig. 5: Response latency distribution ($n = 169$ exchanges).

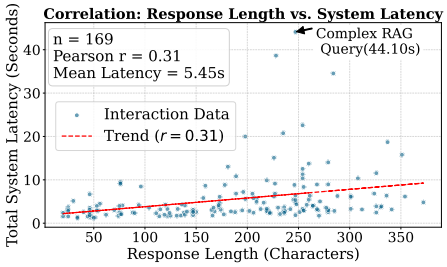


Fig. 6: Response Length vs. Latency ($n = 169$ exchanges).

To understand this delay's source, we analysed the relationship between response length (character count) and total processing latency, as shown in Fig. 6. Following established HRI methodologies for identifying system bottlenecks and performance constraints [25,13], we conducted a Pearson correlation analysis (r) to measure the strength and direction of the linear relationship between different constructs, obtaining $r = 0.31$. We then calculated the coefficient of determination (R^2), which indicates how much of the variation in total system latency can be explained by the length of the generated response. The resulting value of $R^2 = 0.096$ shows that response length explains less than 10% of the variance in total latency. This result suggests that the robot's Text-to-Speech (TTS) engine is not the primary bottleneck affecting its responsiveness. Instead, latency is more likely dominated by other computational overheads, such as API handshakes, context retrieval, and Time-to-First-Token (TTFT), which introduce delays that the conversational framework must accommodate.

6.4 Robot Back-Channelling Behaviours for Latency Compensation

Fig. 7 presents the evaluation results quantifying users' perception of the robot's individual aspects during interaction. The green column on the right shows the overall SUS score across all participants (M = 77.94, SD = 17.55). We further analysed the relationships between these constructs by constructing a correlation matrix. Fig. 8 illustrates the resulting Pearson correlation matrix.

The results presented in Fig. 7 show that the robot achieved a mean *SUS* score of 77.94 (SD = 17.55), placing it in the "Good" to "Excellent" usability tier [6]. Although the measured system latency is around 3.31s, as discussed in Sec. 6.3, participants generally considered the robot response time acceptable. This user perception is reflected in the high rating for *Response Speed* (M = 4.38, SD = 0.81). Furthermore, Fig. 8 suggests only a weak correlation between *Response Speed* and the *SUS* representing the overall system usability.

This result may be explained by the robot's back-channelling behaviours. Specifically, when a user finishes speaking, the robot immediately enters a visible "Thinking" state, expressed through facial animations and gestures that

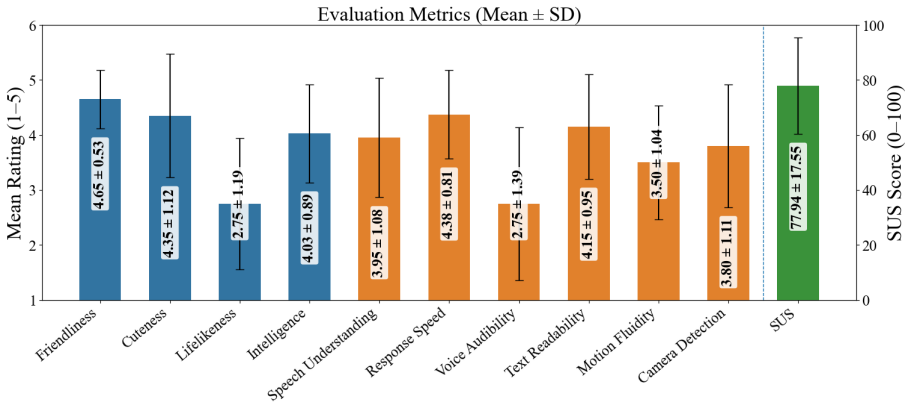


Fig. 7: Users' interaction experience with the robot's functions. The green column on the right side illustrates the overall SUS score ($n = 40$).

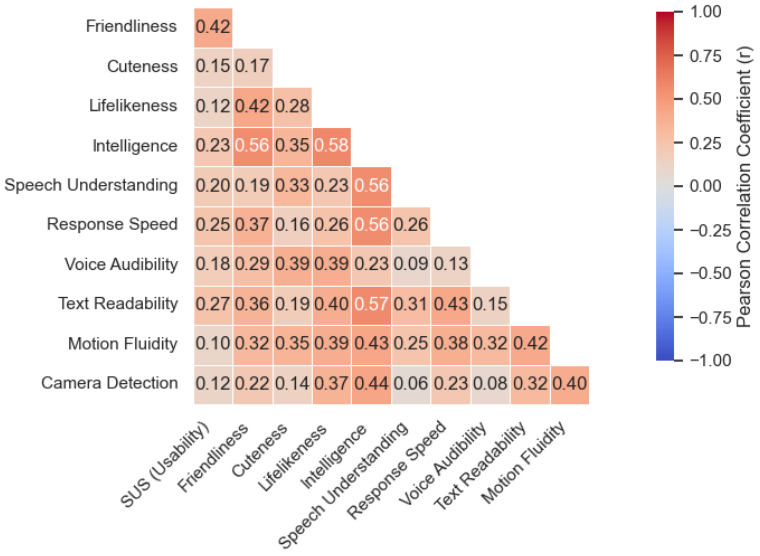


Fig. 8: Pearson correlation matrix of subjective questionnaire metrics ($n = 40$).

signal their ongoing processing. Rather than experiencing a silent pause, users receive such continuous non-verbal feedback indicating that the system has received and is processing their query, which helps maintain engagement. This observation aligns with the self-descriptiveness principle [15], which suggests that when interactive systems experience latency, providing transparent cues about the system's current state can be more effective for maintaining user trust than optimising inference speed alone.

6.5 The Illusion of Competence: Fluency and Warmth

The results presented in Fig. 8 reveal that in our correlation matrix, *Intelligence* is most strongly correlated with *Lifelikeness* ($r = 0.58$), followed by the correlation between *Intelligence* and *Text Readability* ($r = 0.57$), *Intelligence* and *Speech Understanding* ($r = 0.56$), *Intelligence* and *Response Speed* ($r = 0.56$), and notably, *Intelligence* and *Friendliness* ($r = 0.56$).

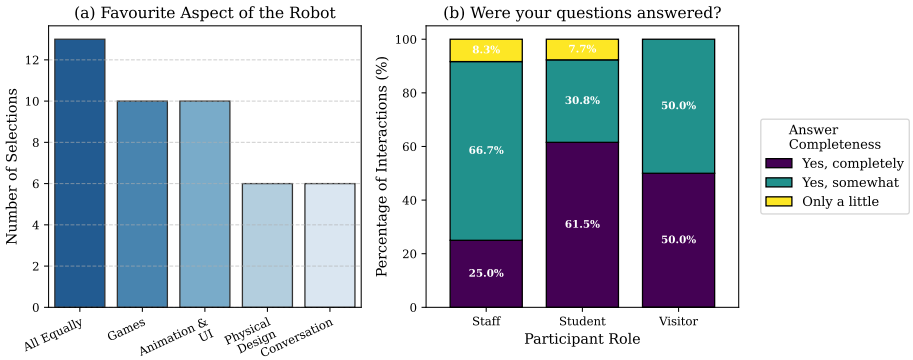


Fig. 9: Questionnaire results ($n = 40$). **(a)** Participants' favourite aspects of the interaction; data reflects 45 total selections due to multi-select permissions. **(b)** User evaluation of the robot's answer.

The strong correlation between *Intelligence* and *Friendliness* suggests a halo effect [16], where perceptions of the robot's social warmth may subconsciously influence users' judgments of its technical competence. This observation is further supported by the categorical results shown in Fig. 9(a), where participants were asked which aspects of the robot they liked most. The responses indicate a clear preference for non-technical features (the robot's entertaining games, facial animations & UI) over its conversational capabilities. Participants' comments also reflect this perception, including feedback such as "Very fun to talk with... Very creative" (Participant 39) and "I like its face + interaction + chatting to it was great" (Participant 11). This finding aligns with prior work suggesting that even an LLM with basic natural language capabilities, when combined with a friendly interface, can create a convincing illusion of competence [37].

6.6 Stylish Design for Managing User Expectations

The strongest predictor of overall satisfaction (SUS) in our matrix, shown in Fig. 8, is *Friendliness* ($r = 0.42$). Interestingly, the lowest-rated dimension is *Lifelikeness* (2.75), as illustrated in Fig. 7.

Lifelikeness refers to the degree to which users perceive the system as alive or anthropomorphic [3]. The relatively low score therefore suggests that participants

tended to perceive the robot as a synthetic agent rather than a human-like entity. Rather than being a limitation, this characteristic can be advantageous. A clearly machine-like appearance helps prevent users from attributing human-level intelligence to the robot, thereby keeping expectations aligned with its actual capabilities [10]. When errors occur, for example, a misheard word or a delayed response, users are more likely to respond with patience rather than frustration, as the robot does not attempt to present itself as human-like or more capable than it actually is. In this context, high *Friendliness* and low *Lifelikeness* complement each other: warmth encourages user engagement, while a clearly stylish design helps maintain user expectations for the robot's performance.

6.7 Divergent Perception on Robot Value

Fig. 9(b) shows that staff and students evaluated the robot's question-answering capabilities differently. In particular, staff members ($n = 12$) gave lower ratings to the robot's answers compared to students. This difference may be explained by staff's deeper familiarity with university-related information, which enables them to more easily identify gaps or inaccuracies in the robot's responses. Yet acceptance remained high, and many staff participants suggested that the robot could serve as an effective triage filter for handling frequently repeated queries. For example, participants noted that the robot would be useful for managing "quick questions that cause queues in busy periods" (Participant 30) and for addressing "easy fixes or identifying common errors before I contact the main iSolutions Team" (Participant 3). In this way, the robot can help filter basic requests, allowing human staff to focus on more complex troubleshooting tasks.

Conversely, students ($n = 26$) evaluated the robot through the lens of *social friction*. In university environments, in several circumstances, students may hesitate to ask basic questions due to concerns about appearing uninformed or being judged by others [20]. Based on participant feedback, many students praised the robot as a low-stakes, judgement-free alternative for "simple/quicker questions that [I] don't want to bother people with" (Participant 32), and for helping ensure that "students could get their homework done before interacting with the team" (Participant 37). This observation aligns with prior research [4], which shows that educational robots and artificial agents lack social judgement, making them effective intermediaries for users who may fear negative evaluation.

These two mechanisms align with the framework [38]. The robot gains staff acceptance by reducing operational workload, while improving student satisfaction by lowering social pressure during help-seeking-two distinct benefits arising from a single deployment.

6.8 Findings

Firstly, our field experiment conducted at the Techhub suggests that the robot's latency is largely dominated by fixed computational overheads, such as API handshakes, context retrieval, and TTFT, which introduce delays that the conversational framework must accommodate. To mitigate the perceived impact of

this latency, the robot's verbal and non-verbal cues should be triggered immediately after the user finishes speaking, thereby decoupling the user's perception of responsiveness from the system's actual processing time. This finding aligns with prior work [5], which highlights that making a robot's internal state readable to users is a key factor for maintaining trust. In this context, the goal is not necessarily to eliminate the pause, but to fill it with clear social signals that sustain user engagement during system processing.

Secondly, our findings suggest that users tend to evaluate a robot's intelligence based on its surface-level interaction capabilities, particularly *Speech Understanding* and *Friendliness*. Consistent with the "Scarecrow" paradigm [37], the immediate value of an embodied agent lies in functioning as a fluent and approachable interface that effectively parses natural language. When the robot accurately recognises user input and responds with social warmth, users are more likely to tolerate minor inaccuracies in its responses. Therefore, when designing LLM-based social robots for university service settings, it is important to prioritise accurate user input recognition and engaging multimodal interaction capabilities, enabling smoother and more natural HRI.

Finally, different perceptions between staff and students regarding the robot's value suggest a unified design implication. A stylised, non-humanoid appearance helps manage user expectations about the robot's capabilities. Users tend to approach the system as a self-service helpdesk support rather than a human expert, which aligns with its intended role: handling basic queries so that helpdesk staff can focus on more complex cases. At the same time, for students, the robot's friendly social cues help reduce the anxiety associated with asking "basic" questions, offering a patient, judgement-free and on-demand source of assistance within the university environment.

6.9 Limitations and Future Works

The absence of a disembodied control condition (e.g., a tablet-based kiosk or mobile devices) limits our ability to isolate the specific contribution of physical embodiment. Furthermore, although we evaluated answer completeness using offline RAGAS metrics and subjective user feedback, we did not objectively assess user information retention or downstream task success. As a result, future longitudinal deployments should incorporate non-embodied baseline conditions (e.g., screen-only interaction) to quantify the impact of embodiment, alongside objective task-completion metrics to verify user information retention in such environments.

7 Conclusion

University helpdesks often receive high volumes of routine enquiries. Deploying LLM-driven social robots in this context represents an emerging research direction that can support helpdesk operations by enabling staff to focus on more

complex queries, while the robot provides on-demand assistance with social interaction capabilities. In this study, both quantitative and qualitative evaluations were conducted to examine how the proposed approach supports question answering and to assess users' perceptions of the robot's usability and perceived usefulness in a helpdesk setting.

Altogether, this paper presents an exploratory case study of a low-cost, scalable LLM-driven social robot deployed for front-line university helpdesk support. The robot design, conversational framework, interaction dataset, and empirical findings reported aim to provide insights for the HRI community in the application of LLM-driven social robots in university service environments.

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